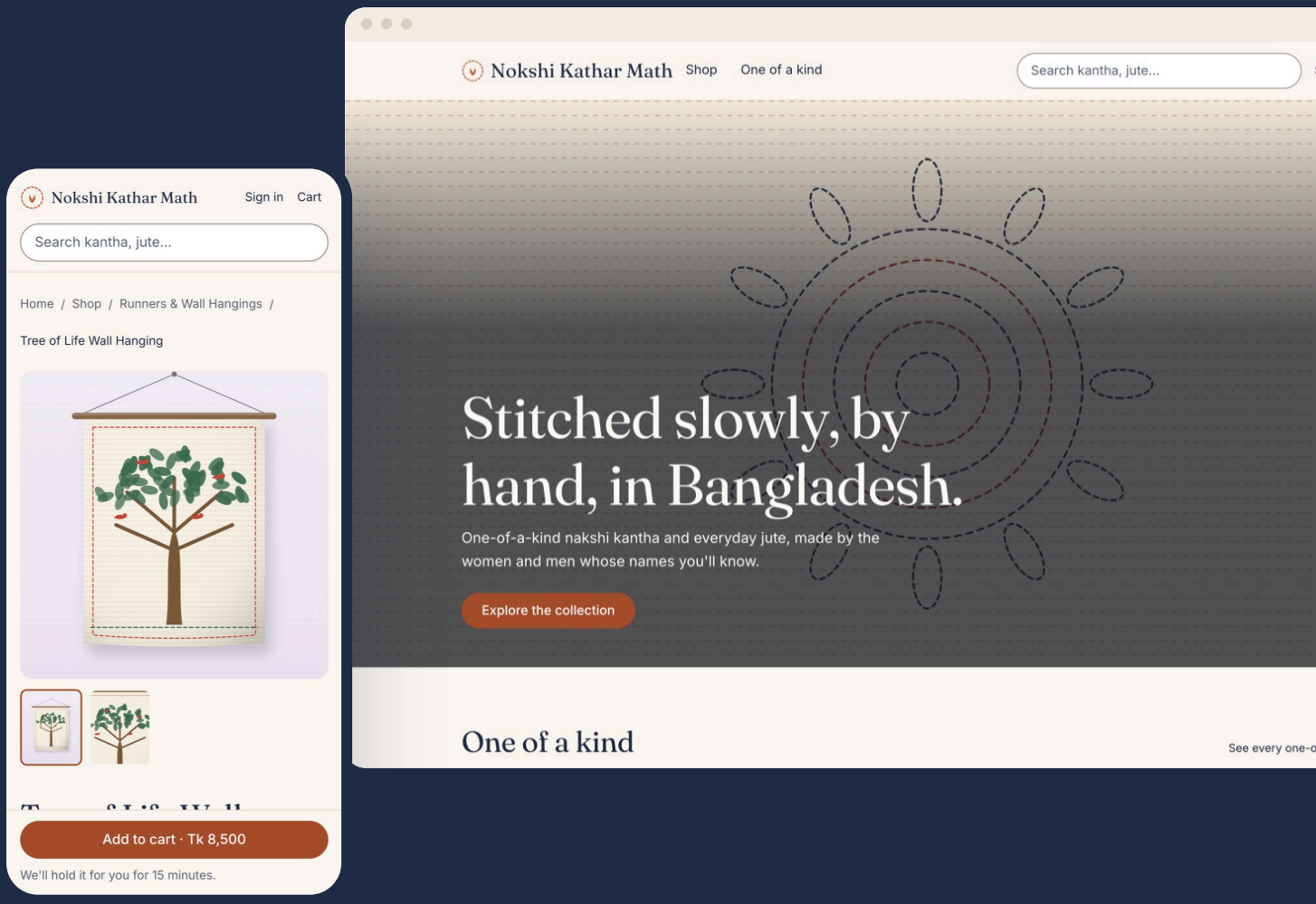
CASE STUDY · FULL-STACK E-COMMERCE

Your custom online store, built from scratch

A walkthrough of Nokshi Kathar Math, a complete shop I designed and built: storefront, checkout, customer accounts and an admin panel.

Angular | ASP.NET Core · .NET 10 | PostgreSQL | Docker | CI on every push



THE PROJECT

A shop for one-of-a-kind handmade goods

The goal

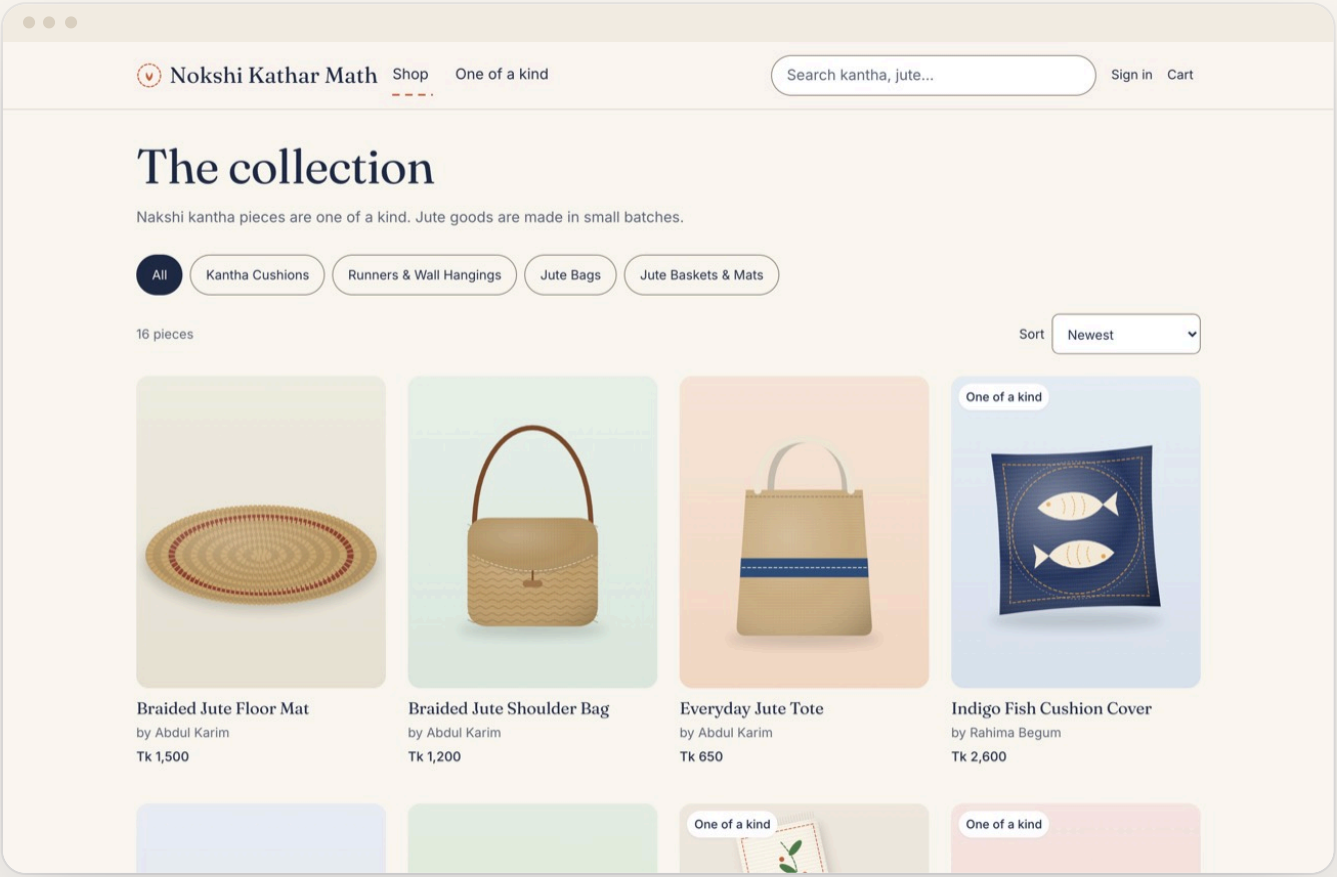
Sell handmade nakshi kantha and jute goods online. Many pieces exist only once, so the shop must never sell the same item twice, even when two shoppers check out at the same moment.

My role

Everything: UX and visual design, Angular frontend, ASP.NET Core API, PostgreSQL database, tests, CI and deployment.

What the shop does

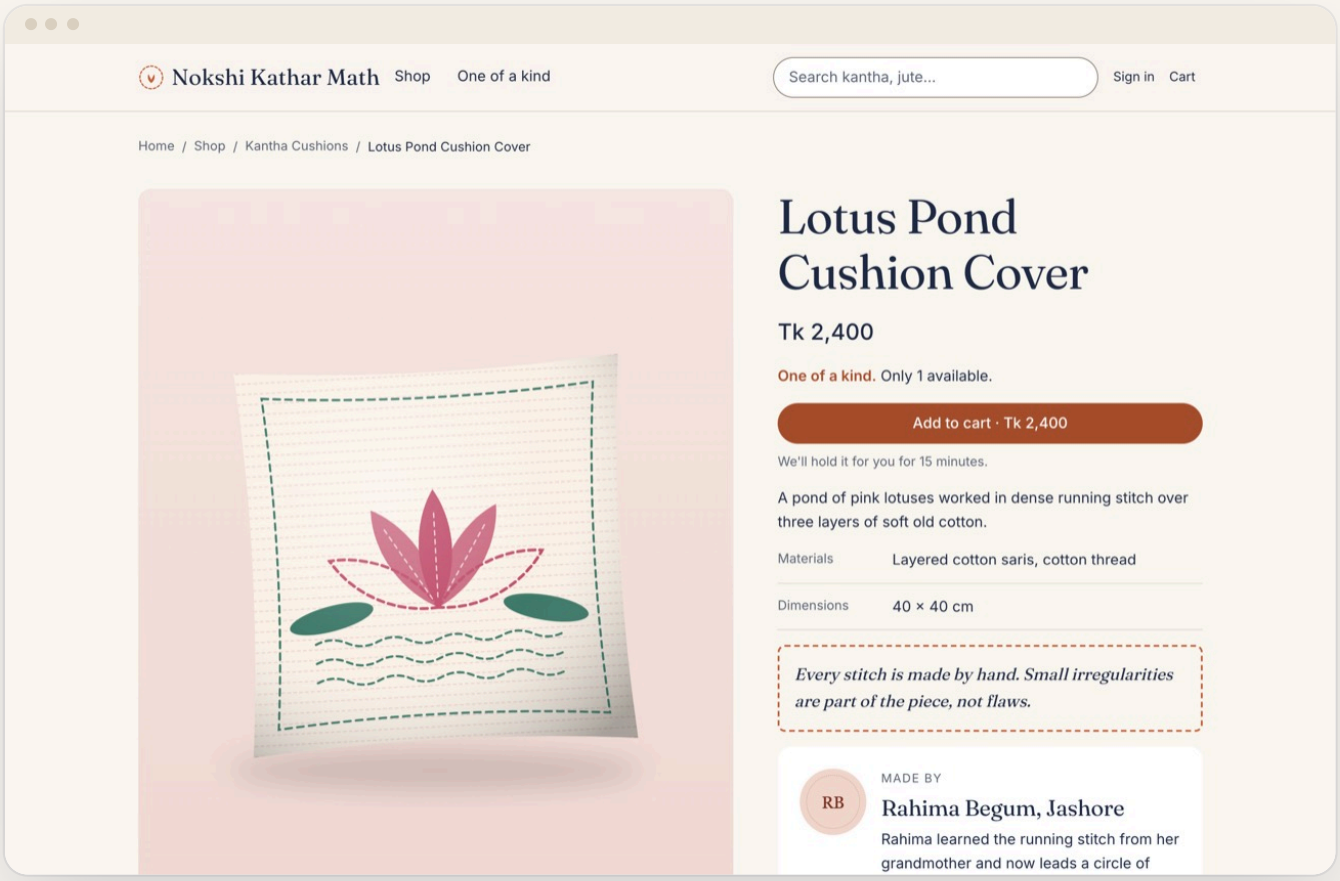
- Catalog with categories, sorting, search and shareable filter links
- One-of-a-kind items are held for 15 minutes when added to a cart
- Cash-on-delivery checkout with prices and stock re-checked on the server
- Customer accounts: the cart built before signing in comes along
- Admin panel for orders, products, stock, photos and artisans



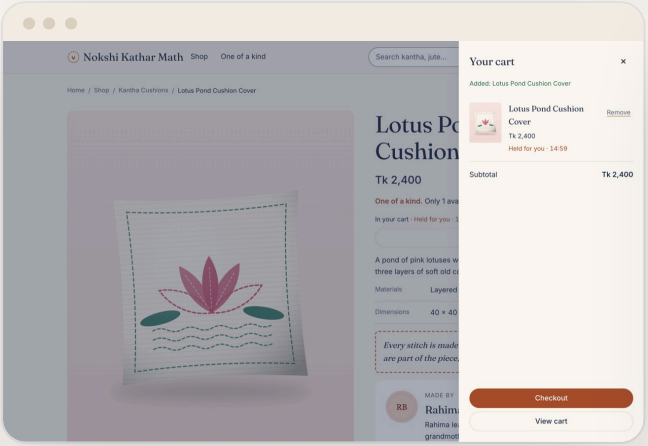
Shop page: filters, sorting and search.

STOREFRONT

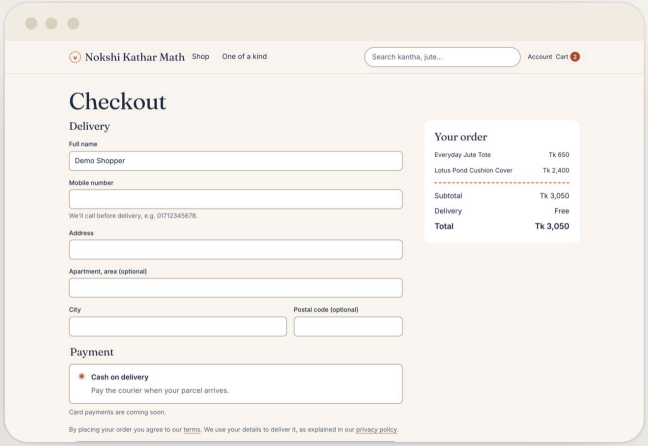
From browsing to checkout



Product page: photos, price, stock status, materials and the maker behind it.



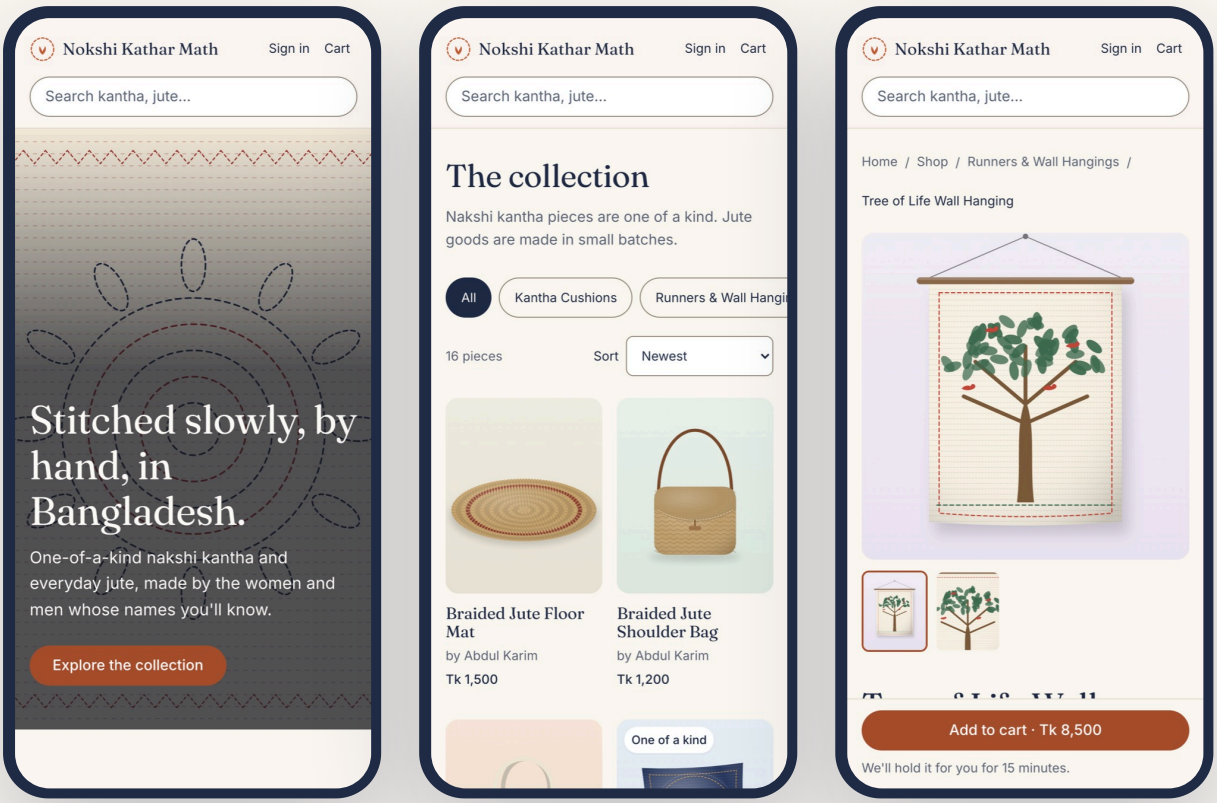
Cart drawer: unique pieces are reserved while the shopper decides.



Checkout: validated delivery details, a short friendly form.

MOBILE-FIRST

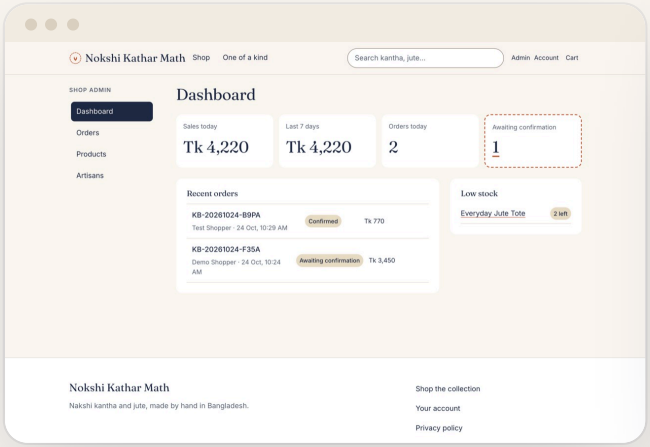
Built for the phone in your customer's hand



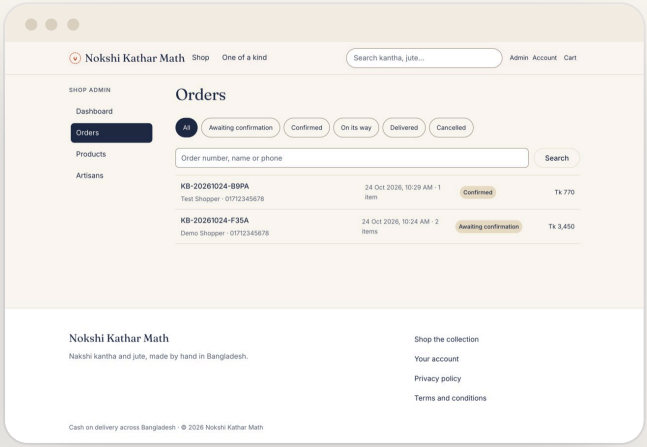
Fast, accessible (WCAG AA contrast, keyboard friendly) with light and dark themes.

ADMIN PANEL

Run the shop without touching code



Dashboard: sales, orders waiting and low-stock alerts.



Orders: status workflow; cancelling restores stock.

UNDER THE HOOD

Production-grade, not a template

No overselling

Holds are a single conditional database update and stock uses optimistic concurrency, so two shoppers can't buy the last item. Both cases have automated tests.

Secure accounts

Short-lived access tokens kept in memory, refresh tokens in httpOnly cookies, hashed and rotated on every use, with reuse detection.

Never trust the client

Prices, holds and stock are re-checked on the server at checkout. Errors come back in one consistent format the UI can explain.

Tested and automated

Integration tests run against a real PostgreSQL database, and the full build, test and lint suite runs on every push.

Architecture

Clean, layered and easy for another developer to pick up. Free-tier friendly hosting.

Angular
Standalone components, signals, lazy routes

ASP.NET Core API
.NET 10, Docker image

PostgreSQL
EF Core, migrations, seed data

Image storage
Signed photo uploads

Multi-store ready: every query is scoped to its shop, so the same code can serve more than one brand.

WORKING TOGETHER

How we'll build your store

01
Discuss
We agree on pages, products, payment method and timeline before you order.

02
Design
A look built around your brand, logo and photos, mobile-first.

03
Build
Storefront, checkout and admin panel, with updates at each milestone.

04
Launch
Deployed to your hosting, full source code and setup notes handed over.

Packages

Starter Store
Catalog, product pages, cart, mobile-ready

Complete Store
Adds search, checkout, customer accounts and hosting setup

Store + Admin
Adds the admin panel for orders, stock and photos, deployed live

Let's build your store

Message me on Fiverr to get started